

BERTH OPTIMIZATION & YARD PREPARATION FRAMEWORK

Operational Planning & Decision Support Document

1. Introduction

In container terminal operations, berth allocation is one of the most critical decisions. Traditionally, berth selection is often based on availability and vessel size. However, this approach does not consider:

- Where the cargo is located in the yard
- Travel distance for internal trucks (ITVs)
- Crane idle time risk
- Yard congestion
- Historical vessel productivity
- Conflict with other working vessels

The purpose of this document is to define a structured operational method to:

1. Identify the most suitable berth for an upcoming vessel
2. Predict expected port stay time
3. Detect operational conflict risks
4. Learn from previous vessel delays
5. Prepare the yard strategically before vessel arrival

The goal is simple:

Place the vessel in the best berth to minimize travel, reduce congestion, and complete operations faster.

2. Berth Selection Based on Yard Cargo Distribution

2.1 Operational Principle

When a vessel is scheduled to arrive, most of its loading containers are already inside the yard.

If the majority of these containers are located near a specific berth zone, placing the vessel at that berth will:

- Reduce internal truck travel distance
- Reduce fuel consumption
- Reduce crane waiting time
- Reduce yard congestion
- Improve overall productivity

Therefore, berth selection should consider cargo location concentration.

2.2 Step 1 – Identify Loading Container Locations

Before confirming the berth, the system must:

1. Extract all containers nominated for loading.
2. Identify for each container:
 - Yard block
 - Yard row
 - Yard bay
 - Equipment class
 - Weight class
3. Map each yard block to its closest berth area.

This creates a clear picture of where the cargo is physically located.

2.3 Step 2 – Generate Yard Heat Map

A yard heat map must be created to visually represent container concentration.

The heat map should:

- Highlight blocks with high concentration (Red)
- Medium concentration (Orange)

- Low concentration (Green)

This visual tool allows planners to quickly see where the vessel's cargo is clustered.

Example:

Terminal has:

- Berth 1
- Berth 2
- Berth 3

Vessel A has 1,500 loading containers.

After analysis:

- 78% located in blocks near Berth 1
- 40% located near Berth 2
- 25% located near Berth 3

Operational interpretation:

If the vessel is berthed at Berth 1:

- Minimal travel
- Lower congestion
- Higher crane efficiency

If berthed at Berth 3:

- Long travel distance
- Higher risk of ITV queue
- Increased crane idle time

2.4 Step 3 – Distance Impact Simulation

For each berth option, the system must calculate:

- Total Laden Travel Distance
(Yard Block → Selected Berth × Number of Containers)
- Estimated Unladen Travel
(Return trips, repositioning)
- Corridor congestion impact

This provides measurable data rather than opinion-based decisions.

Final output example:

Berth	Cargo Concentration	Total Travel Distance	Congestion Risk
B1	High (78%)	Low	Low
B2	Medium	Medium	Medium
B3	Low	High	High

Recommended berth = Lowest combined operational impact.

3. Port Stay Time Prediction

3.1 Objective

To predict how long the vessel will remain at berth using historical performance data.

3.2 Operational Logic

Port stay depends on:

- Total moves (Load + Discharge)
- Number of cranes assigned
- Average crane productivity
- Crane intensity (how many cranes work simultaneously)

3.3 Calculation Method

Step 1: Estimate total moves

Total Moves = Planned Load + Planned Discharge

Step 2: Identify historical data for similar vessels:

- Average number of cranes assigned
- Average moves per hour per crane

Step 3: Calculate expected port stay:

Port Stay =

$\text{Total Moves} \div (\text{Number of Cranes} \times \text{Average Productivity})$

Example:

Total moves = 2,000

Cranes assigned = 4

Productivity = 30 moves/hour

$\text{Port Stay} = 2000 \div (4 \times 30) = 16.6 \text{ hours}$

3.4 Operational Benefit

This helps:

- Marine scheduling
- Manpower planning
- Crane allocation
- Berth window planning
- Customer communication

4. Berth Conflict & Crane Clash Analysis

4.1 Objective

To prevent operational conflicts when multiple vessels work simultaneously.

4.2 Operational Risks

Conflicts occur when:

- Two vessels load from same yard blocks
- Heavy ITV traffic in same corridors
- Crane rail overlap

- Equipment competition

These situations increase:

- Crane idle time
- Truck waiting time
- Vessel delay

4.3 Required Analysis

The system must check:

1. Percentage of shared yard blocks
2. Corridor overlap
3. Crane working zone overlap
4. Equipment demand conflicts

Output must classify risk:

- Low risk
- Medium risk
- High risk

And suggest mitigation:

- Change berth
- Adjust crane allocation
- Pre-consolidate yard cargo

5. Historical Delay Root Cause Analysis

5.1 Objective

To understand why previous vessels were delayed and avoid repeating mistakes.

5.2 Operational Checks

For previous delayed vessels, analyze:

- Crane idle time
- Container availability timing
- ITV arrival delays
- High M-cycle percentage
- Poor stacking strategy
- Long unladen travel
- Crane crossing conflicts

5.3 Interpretation

If delay reason = Container not ready
→ Improve yard readiness before arrival

If delay reason = High M-cycle
→ Improve stacking and consolidation

If delay reason = ITV congestion
→ Review corridor planning

Learning from history improves predictability.

6. Yard Preparation & Weight Distribution Strategy

6.1 Objective

To prepare the yard before vessel arrival to reduce reshuffles and crane delay.

6.2 Historical Stowage Pattern Analysis

The system must analyze:

- Weight distribution loaded above deck

- Weight distribution loaded below deck
- Equipment class distribution
- Discharge port grouping

6.3 Yard Strategy

Based on prediction:

- Heavy containers positioned close to berth
- Light containers positioned for upper stack loading
- Group containers by discharge sequence
- Reduce reshuffle requirement

Expected result:

- Lower reshuffle %
- Better dual cycle
- Faster crane execution

7. Dynamic Recalculation Based on ETA Accuracy

If vessel ETA changes:

- Re-run berth simulation
- Recalculate congestion
- Update port stay prediction

This ensures berth plan remains optimized in real time.

8. Overall Operational Benefits

Implementing this framework will:

- Reduce total travel distance

- Reduce fuel cost
- Reduce crane idle time
- Improve vessel turnaround time
- Improve berth utilization
- Reduce operational uncertainty
- Support data-driven decision making

9. Final Conclusion

Berth selection must move from availability-based decision to data-driven operational optimization.

This framework ensures that berth allocation considers:

- Cargo concentration
- Travel distance
- Crane productivity
- Conflict risk
- Historical delays
- Yard readiness

By following this structured operational approach, the terminal can achieve:

- Faster vessel completion
- Reduced congestion
- Predictable performance
- Continuous improvement